

Microeconomic Analysis

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Homework 1

Due Thursday 17th September

Question 1 Some things to clear up from class

1. Come up with a choice correspondence that satisfies property α but not property β , and another that satisfies β but not α
2. Show that if a choice correspondence has a utility representation then it must satisfy α and β
3. Prove that a choice correspondence satisfies properties α and β if and only if it satisfies WARP
4. Here is an alternative definition of WARP, which we will call WARP*

Axiom 1 (WARP*) *For any set S , there exists a b^* in S such that, for any T including b^* , if $C(T) \in S$ then $C(T) = \{b^*\}$*

Show that, if we restrict attention to choice functions (i.e. C is single valued) a data set satisfies WARP* if and only if it satisfies WARP. Is this still the case if we allow for C to be a choice correspondence?

Question 2 Utility maximization is not the only choice procedure that are consistent with α and β . There are also other choice procedures that will satisfy these conditions and so are indistinguishable from rational choice. Consider the following decision making procedures. Prove whether or not they will result in choices that satisfy α and β (Feel free to assume that there are a finite number of alternatives in each case). If they are consistent with the

α and β , and therefore admit a utility representation, discuss the relationship between such representations and their ‘true’ utility (if the decision making model has such a thing)

1. Satisficing: A decision maker (DM) is choosing between books from a set B . They have a utility function $u : B \rightarrow \mathbb{R}$, and a ‘threshold utility’ level u^* . In any choice set, they search through the books alphabetically by title, and choose the first book that has utility level u that is equal to or above u^* . If they have not found any such book by the time they reach the end of the choice set, they will choose the book with the highest utility (to make things simpler you can assume that there is no indifference - i.e. no two books have the same utilities)
2. Elimination by Aspects: The DM is choosing between kettles. Each kettle is defined by its price, its energy efficiency (on a scale from 1 to 5) and its attractiveness (on a scale of 1 to 3). When given a choice set the DM
 - (a) Selects all the kettles with the highest attractiveness rating
 - (b) From those kettles, selects those with the best energy efficiency
 - (c) From those that remain, they choose the cheapest
3. The DM ranks the alternatives according to price, then looks at the three cheapest and chooses the one with the highest utility
4. The DM has a true utility function for kettles based on price, energy efficiency and attractiveness. However, to save time, they buy they always just buy the cheapest kettle.
5. The DM ranks the alternatives according to a utility function, and in any choice set chooses all median elements (note that if the median has more than one element all are chosen)
6. Rational Shortlisting: The DM is choosing between routes to drive home. For some routes they have live transit information and for some they do not. The DM makes decisions in two stages. First, they remove any routes such that there is a quicker available route according to the live transit information. Of the remaining routes they choose the shortest (in miles)

Question 3 Let $\succeq$ be a complete relation on a non-empty set X , and for any $S \subset X$ finite define

$$C_{\succeq}(S) = \{x \in S | x \succeq y \text{ all } y \in S\}$$

1. Show that if $\succeq$ is transitive $C(S)$ is non-empty
2. We say a binary relation is acyclic if there is no k and set $x_1, \dots, x_k \in X$ st $x_1 \succ x_2 \succ \dots \succ x_k \succ x_1$. Show that acyclicity is a strictly weaker property than transitivity
3. Show that if $\succeq$ is acyclic $C(S)$ is non-empty
4. Show that if $C(S)$ is non empty for every finite S , then $\succeq$ is acyclic

Question 4 One popular model of choice behavior in the marketing literature is choice with consideration sets. The basic idea is that, when presented with a choice set A , the decision maker only really thinks about a subset of those alternatives - $E(A)$ - and chooses the best of those

Definition 1 We say a set of choice data can be explained as choice with consideration sets if there is (i) a utility function $u : X \rightarrow \mathbb{R}$ and (ii) a consideration set correspondence $E : 2^X / \emptyset \rightarrow 2^X / \emptyset$ such that $E(A) \subseteq A$ and

$$C(A) = \arg \max_{x \in E(A)} u(x)$$

In other words, for each set A , $E(A)$ defines the set of alternatives that the decision maker considers. They then choose the best option from $E(A)$ according to u .

For simplicity, let's assume that we are dealing with choice functions (not correspondences) and that there is no indifference, and that X is finite.

1. Show that, without further restrictions on E , a model of choice from consideration sets can explain **any** choice function
2. For the remainder of the question, add the restriction

$$\text{If } x \in B \subset A \text{ then } x \in E(A) \text{ implies } x \in E(B)$$

This means that consideration gets harder in larger choice sets, so if x is considered in the bigger set A then it should also be considered in the smaller set B

Is this restriction implied by each of the following choice procedures?

- (a) In each choice set consider the three cheapest cars (or all the cars if there are fewer than three)

- (b) In each choice set consider the fastest car and the car with the best safety rating
 - (c) In each choice set consider cars that are unusual: You count up the number of cars in the choice set which are green, red, brown etc and consider only those that are of the least common color
3. Is the following set of choices consistent with this model?

$$C(\{x, y, z\}) = x$$

$$C(\{x, y\}) = y$$

- 4. Show that the behavior in part 3 implies that $u(y) > u(x)$ if the model is correct
- 5. Show that this model does not imply WARP.
- 6. Consider the following modification of the WARP* axiom of question 1

Axiom 2 *For any set S , there exists a b^* in S such that, for any T including b^* , if $C(T) \in S$ and $b^* = C(T')$ for some $T' \supset T$ then $C(T) = b^*$*

Show that the consideration set model described above does imply this behavior

- 7. Consider the following variant of the above model. Rather than the condition in part 2, we have the condition

$$E(A \setminus x) = E(A) \text{ if } x \notin E(A)$$

In other words if x is not considered at A , then removing it does not change the consideration set. Does the resulting model still satisfy the axiom from part 6?